**Senior Foot Care**
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Foot problems affect an estimated 71–87% of adults over the age of 65, yet they remain chronically under-reported and under-treated in primary care settings. In community-dwelling older adults, foot pain is among the top five musculoskeletal complaints, with prevalence increasing steadily with age.

The consequences of unmanaged foot pathology extend well beyond discomfort:

- Falls: Foot pain, reduced plantar sensation, and altered gait mechanics collectively account for a significant proportion of fall incidents in older adults. Studies indicate that individuals with foot pain are twice as likely to experience a fall compared to pain-free peers.
- Functional decline: Reduced walking speed, restricted community mobility, and activity avoidance are well-documented sequelae of chronic foot pain.
- Hospitalization: Complications arising from neglected foot wounds — particularly in diabetic patients — represent a leading cause of non-traumatic lower-limb amputation globally.
- Psychosocial impact: Chronic foot pain correlates significantly with depression, social isolation, and reduced self-rated health in longitudinal cohort studies.

## Age-Related Changes in the Foot

| Structure | Age-Related Change | Clinical Consequence |
|---|---|---|
| Fat pad (heel & forefoot) | Atrophy and reduced elasticity | Increased plantar pressure and pain |
| Ligaments and tendons | Loss of collagen cross-linking, reduced tensile strength | Arch collapse, flatfoot deformity |
| Joints | Cartilage thinning, osteophyte formation | Hallux rigidus, reduced ankle dorsiflexion |
| Toe alignment | Progressive deformity from intrinsic muscle wasting | Hammer toes, claw toes, hallux valgus |
| Arch | Medial longitudinal arch descent | Pes planus, metatarsalgia |

### *Neurological Changes*

Peripheral sensory nerve conduction velocity declines with age, resulting in:

- Reduced vibration and proprioceptive sensitivity — impairing balance and gait stability
- Diminished protective pain sensation — allowing minor injuries to progress undetected
- Impaired thermoregulation — increasing susceptibility to burns and frostbite

### *Vascular Changes*

Peripheral arterial disease (PAD) prevalence increases sharply after age 65, affecting approximately 12–20% of this cohort. Even in the absence of frank PAD, capillary density and microcirculatory efficiency decline, resulting in:

- Slower wound healing
- Dry, fragile skin prone to fissuring
- Increased nail and skin infection susceptibility

### *Dermatological Changes*

- Skin thins due to reduced dermal collagen and epidermal turnover
- Sebaceous and sweat gland activity diminishes, causing xerosis (pathological dryness)
- Nail plates thicken, become brittle, and are increasingly prone to fungal colonization (onychomycosis)
- Callus and corn formation accelerates under pressure points

Common Foot Conditions in Older Adults

*Hallux Valgus (Bunion)*
A lateral deviation of the great toe at the metatarsophalangeal joint, hallux valgus affects up to 36% of adults over 65. It is strongly associated with ill-fitting footwear over decades. Management includes wide toe-box footwear, orthotic correction, and in severe cases, surgical realignment.

*Plantar Fasciitis*
Degeneration and microtearing of the plantar fascia at its calcaneal insertion produces classic heel pain on first rising in the morning. Management includes calf and plantar stretching, night splints, arch supports, and activity modification.

*Metatarsalgia*
Forefoot pain from fat pad atrophy and increased plantar pressure over the metatarsal heads. Addressed with metatarsal pads, cushioned footwear, and offloading insoles.

*Hammer Toes and Claw Toes*
Progressive flexion contractures of the interphalangeal joints cause dorsal pressure lesions and corns. Padding, toe spacers, and wide deep toe-box shoes reduce symptoms; severe cases require orthopaedic review.

*Onychomycosis (Fungal Nail Infection)*
Affects up to 50% of adults over 70. Characterized by thickened, discolored, crumbling nails. Topical and systemic antifungals are first-line treatments; mechanical debridement by a podiatrist improves treatment penetration.

*Xerosis and Heel Fissures*
Dry skin, especially at the heels, can crack deeply enough to create entry wounds for bacteria. Daily emollient use is the cornerstone of prevention.

*Peripheral Neuropathy*
In diabetic and non-diabetic patients alike, peripheral neuropathy creates a high-risk "silent injury" environment where blisters, ulcers, and infections progress without pain warning. Vigilant daily inspection is mandatory.

Daily Foot Care Habits

*Daily Inspection*

1. Conduct foot inspection in good lighting, preferably with a magnifying mirror for those with limited flexibility.
2. Examine all surfaces: dorsum, plantar aspect, inter-digit spaces, nail margins.
3. Look specifically for: blisters, cuts, abrasions, redness, swelling, discoloration, or nail changes.
4. For individuals with diabetic neuropathy or visual impairment, a caregiver or family member should perform or assist with inspection daily.

The absence of pain does not indicate the absence of injury in neuropathic patients. Systematic visual inspection is non-negotiable.

*Washing*

- Wash feet daily with lukewarm water (not hot — test with elbow or thermometer if sensation is reduced).
- Use a mild, pH-balanced soap; avoid harsh antibacterial soaps that strip protective skin flora.
- Soaking beyond 5 minutes is not recommended, as prolonged immersion macerate and softens skin, increasing fissure risk.
- Dry thoroughly, especially between the toes, using a soft towel with gentle patting (not rubbing). Retained moisture fosters fungal infection.

*Moisturising*

- Apply an urea-based emollient (10–25% urea) or a lanolin/shea butter formulation daily to all foot surfaces except inter-digit spaces.
- Apply after washing while skin is slightly warm.
- Avoid applying between the toes — excess moisture here promotes maceration and fungal growth.
- For severe xerosis, occluded overnight application (socks over cream) significantly accelerates skin repair.

*Nail Care*

- Trim nails straight across — never cut into the corners, which predisposes to ingrown toenails.
- Use clean, sharp nail clippers or nail scissors appropriate for the nail thickness.
- File sharp edges with an emery board after cutting.
- Optimal nail length: level with the tip of the toe, neither protruding nor cut short into the nail bed.
- If nails are excessively thickened (onychogryphosis) or fungal, refer to a podiatrist for safe debridement rather than attempting home cutting.

*Callus and Corn Management*

- Do not attempt self-removal with blades, corn-plasters containing salicylic acid (in diabetic/neuropathic patients), or sharp instruments. These carry significant ulceration risk.
- Gentle pumice stone use after bathing softens callus; cease if skin reddens.
- Podiatric debridement every 6–12 weeks is appropriate for established callus.

## Footwear Selection Guidelines

Footwear is among the most modifiable risk factors for foot pain, skin injury, and falls in older adults. Studies consistently demonstrate that a significant proportion of older adults wear shoes that are either too short, too narrow, or excessively worn.

| Feature | Recommendation | Rationale |
|---|---|---|
| Length | 1–1.5 cm (≈ thumb's width) between longest toe and shoe end | Prevents nail trauma; accommodates foot length changes during ambulation |
| Width | Widest part of the shoe matches widest part of the foot | Prevents lateral pressure, bunion aggravation, and corn formation |
| Toe box | Roomy, non-tapering, sufficient depth | Accommodates toe deformities; prevents dorsal pressure lesions |
| Heel counter | Firm and rigid | Stabilises the calcaneus; reduces pronation and ankle instability |
| Sole | Firm midsole, textured non-slip outsole | Energy return; fall prevention on slippery surfaces |
| Cushioning | Moderate forefoot and heel cushioning | Compensates for fat pad atrophy |
| Fastenings | Laces, hook-and-loop (Velcro), or | Accommodates fluctuating oedema; ensures secure fit |

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|---|---|---|
| | adjustable buckle | |
| Heel height | Ideally 1.5–2.5 cm (low block heel) | Reduces Achilles tendon strain without destabilising gait |
| Weight | Lightweight construction | Reduces fatigue and trip risk from foot-dragging |

*Footwear Styles to Prioritise*

- Walking shoes / athletic trainers with a wide toe box, cushioned midsole, and lace or Velcro closure — the most broadly protective option
- Therapeutic/extra-depth shoes (available by prescription in many health systems) for those with significant deformity, neuropathy, or orthotics
- Adjustable sandals with heel strap and contoured footbed — suitable for indoor summer use; a heel strap is mandatory for stability
- Mary-Jane style shoes with broad toe box and ankle strap for those who prefer traditional styling

To Avoid:

- High heels (>2.5 cm): Shift forefoot loading, reduce ankle stability, accelerate hallux valgus and metatarsalgia
- Narrow, pointed toe-box shoes: Compress toes and accelerate deformity
- Flat, unsupported shoes (ballet flats, flip-flops): Provide no arch support or heel stabilization; increase plantar fasciitis risk
- Shoes without heel backs (mules, backless slippers): Require abnormal toe-gripping to retain; increase trip risk
- Worn-down shoes: Asymmetric sole wear alters gait mechanics and increases fall risk

*Shopping Advice*

- Shop for shoes in the afternoon or after a moderate walk, when feet are at their largest (feet swell approximately 5–8% during the day).
- Have both feet measured at each purchase — foot size commonly changes with age.
- Try on both shoes and walk on a hard surface, not just carpet, to assess fit dynamically.
- Bring any custom orthotics to ensure adequate depth.
- Replace footwear when the outer sole shows uneven wear, the midsole feels compressed, or the heel counter collapses — typically every 12–18 months for daily-wear shoes.

*Indoor and Overnight Footwear*

- Never walk barefoot, even indoors — on hard floors this increases plantar pressure, and underfoot hazards cause lacerations that heal poorly in elderly skin.
- Use supportive slippers with a firm sole, enclosed heel, and non-slip sole.
- Avoid soft moccasin-type slippers that provide no support and can slip off.

## Orthotic Devices and Assistive Insoles

*Over-the-Counter Insoles*

Prefabricated insoles with arch support, metatarsal pads, or heel cups provide meaningful symptomatic benefit for plantar fasciitis, metatarsalgia, and mild flatfoot in many patients. They are an appropriate first-line adjunct before custom orthotic referral.

*Custom Foot Orthotics*
Prescribed and fabricated by a podiatrist or orthotist following biomechanical assessment and gait analysis. Indicated for:

- Significant structural foot deformity
- Failed conservative management
- Diabetic foot offloading
- Post-surgical rehabilitation
- Pronounced limb length discrepancy

*Toe Spacers and Padding*
Silicone toe separators reduce inter-digital pressure and friction in hallux valgus and overlapping toes. Felt padding around corns and calluses offloads pressure temporarily, though underlying cause must be addressed.